APPENDIX A

Running the Visualizations

IN THIS APPENDIX A

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We believe that active learning is the best way to master new material. Reading a book and attending a lecture provide the key concepts, but applying that knowledge by solving practice problems and performing experiments makes the abstract ideas more concrete.

To understand data structures and algorithms, it helps to see them in action. To that end, we have developed interactive visualizations of most of the data structures and algorithms described in this book. These programs show the step-by-step processing of data in each structure, allowing you to see all the key operations. You can try the structures with your own data to see how they handle it. The individual chapters suggest exercises for you to try with the visualizations to ensure that you see the important behaviors of each structure and algorithm.

There are several ways for you to run the visualization programs. The best way for each reader depends on how much programming you plan to do. For readers who plan to or already work as software developers, we encourage you to download the source code and run the programs on a computer with an editor or other integrated development environment. You can even help future students by contributing to the open-source software project for the visualizations. Your experience can help suggest new ways of making the complex ideas easier to understand by others. Follow the instructions in "For Developers: Running and Changing the Visualizations."

If you don't plan to do much programming but still want to try running the visualizations on your own data, you can download them and run them on a personal computer.

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This approach would be a good choice for someone who manages or analyzes software development, without doing much coding on their own. If that suits your needs, follow the instructions in "For Managers: Downloading and Running the Visualizations."

If you don't plan on doing any programming and have a connection to the Internet, you can run the visualizations in a web browser. The programs were written in Python and display their visualization using Tk, which means they don't execute in a browser. That adds extra layers of interface that somewhat limits what you can do and see. For someone without access to a computer for software development, however, this may be the only option. If that's the case, follow the instructions in "For Others: Viewing the Visualizations on the Internet."

After you have been able to launch the visualizations, make sure to read the section "Using the Visualizations" at the end. This section explains some of the key features of all the visualizations and their user interfaces, including some limitations in the visualization that are not necessarily present in actual implementations of the data structures.

For Developers: Running and Changing the Visualizations

To maximize your ability to run the programs and modify them for your own experiments or to contribute to the open-source software project, you need at least three pieces of software on the computer where you plan to do the development:

- Python
- Git
- The visualization software

The first two come in different versions depending on your computer's operating system.

Getting Python

To get the Python programming language, you have many options. First, check whether it is already available on your system. If you're already familiar with command-line interfaces like the Windows Power Shell, the Windows Command interpreter, or macOS and Linux Terminals, try launching one of those interfaces and typing the command:

```
python3
```

If Python release 3 is installed, this should produce some lines that look something like:

```
Python 3.7.5 (default, Nov  4 2019, 10:26:32)
[Clang 9.0.0 (clang-900.0.39.2)] on darwin
Type "help", "copyright", "credits" or "license" for more information.
>>>
```

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If this code doesn't show up, then you probably need to install Python. There are many places on the Internet to get Python. The primary one, www.python.org/downloads, has all the releases of Python and many preconfigured downloads for the different operating systems. It's also the official source for all information about the programming language, the different releases, documentation, and proposed changes for the future. The visualizations were made using Python release 3. You need a version that starts with "3."; do not use one that starts with "2." or lower. If this is your first time using Python, you may want to follow the documentation links to read the guides for beginners.

Getting Git

The second tool, git, manages collections of software and their different versions. It's exceptionally well suited to managing source code because it provides many ways to track the changes in text files. Git became the tool of choice for the developers of the Linux operating system. One of its key features is the distributed nature of the software collections, called *repositories*, that it manages. While github.com maintains a copy of most of these repositories, every developer gets their own complete repository including all the history of changes, in their local copy of the repository.

The `git` command-line tool is incredibly powerful, and not surprisingly, incredibly complex. It was invented by software professionals for use by software professionals. Its complexity, however, can be a hindrance for newcomers. Many users just need the basics, and graphical user interfaces can make that much easier. The GitHub Desktop tool provides a straightforward graphical interface to working with repositories, their branches, and commits (changes) to those branches.

Learning how to use the `git` command-line tool is beyond the scope of this book. Instead, we describe how to use GitHub Desktop. If you're already familiar with the `git` command-line tool, however, go ahead and use that, because it can do everything GitHub Desktop does and more.

Use your web browser to visit https://desktop.github.com. It should auto-detect the operating system of your computer and propose a download option. At the time of this writing, only Windows and macOS are supported (on Linux, you need to use the `git` command-line tool, which is often bundled with the Linux distribution). Follow the download instructions to install GitHub Desktop.

Both GitHub Desktop and the `git` command-line tool allow you to review the history of changes in a repository and make new changes to the files it contains. In the next section, we discuss how to get the repository that contains the visualizations for this book. If you plan to make contributions to the open-source visualization project, you also need an account on the GitHub.com website. That is where the project's repository is stored, and your user account allows you to contribute to many different projects. Go to https://github.com and select the Sign up button to create a new account.

Getting the Visualizations

The third piece of software you need is the visualization code itself, which is bundled in a git repository available at https://github.com/JMCanning78/datastructures-visualization.

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The process of downloading a git repository is called *cloning* the repository. If you are using Github Desktop, go to the File > Clone Repository dialog. Choose the GitHub.com tab and then start typing in the repository name, `JMCanning78/datastructures-visualization`. The repository should pop up as a choice that you can select. The dialog lets you specify a Local Path on your computer where it will store the clone. After selecting the repository on GitHub.com and specifying the local path, select the Clone button to make the clone.

If you're using the command-line version of `git`, the best way to get the visualization software is to put one of your command-line windows into the parent directory where you want to store the software and then run the `git clone` command. For example, if you wanted to store the repository in `/Users/Me/myrepositories`, the commands would look something like this:

```
cd /Users/Me/myrepositories
git clone https://github.com/JMCanning78/datastructures-visualization
```

The `cd` command is short for change directory. Your command-line tool might have a different name for the change directory command. The `git clone` command creates a subdirectory called `datastructures-visualization` that contains a clone of the git repository. You can use the `cd` command again to change the current working directory to be that subdirectory. From that directory, you can launch the visualizations by running

```
python3 DatastructureVisualizations.py
```

If you get a message about missing required modules, you may need to add some modules to your Python environment. Try running the Python installer program (`pip3`), as follows:

```
pip3 install -r requuirements.txt
```

If you find that your computer doesn't have `pip`, there is help on ways to install it available at https://pip.pypa.io/en/stable/. When `pip3` runs successfully, try the previous command to run the visualizations.

If you prefer to use an Integrated Development Environment (IDE) for your Python development work, you need to register the cloned repository with the IDE. From there, you should be able to run `DatastructureVisualizations.py` or any other Python programs.

For Managers: Downloading and Running the Visualizations

If you have a macOS or Windows desktop computer, you may be able to download a prepackaged collection of all the visualizations. It runs basically the same program as the DatastructureVisualizations.py described previously. Browse https://datastructures.live/ and look for the links related to downloads.

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The software bundles are cryptographically signed to help ensure the integrity of the software. Your computer and its malware protection software may ask whether you trust the downloaded visualization software before allowing you to run it.

For Others: Viewing the Visualizations on the Internet

You can view the visualizations by visiting https://datastructures.live/view/ in a web browser. When you select the Run button, the website runs the visualizations on a server and copies the output to a pane in your web browser window. The initial screen looks something like Figure A-1.

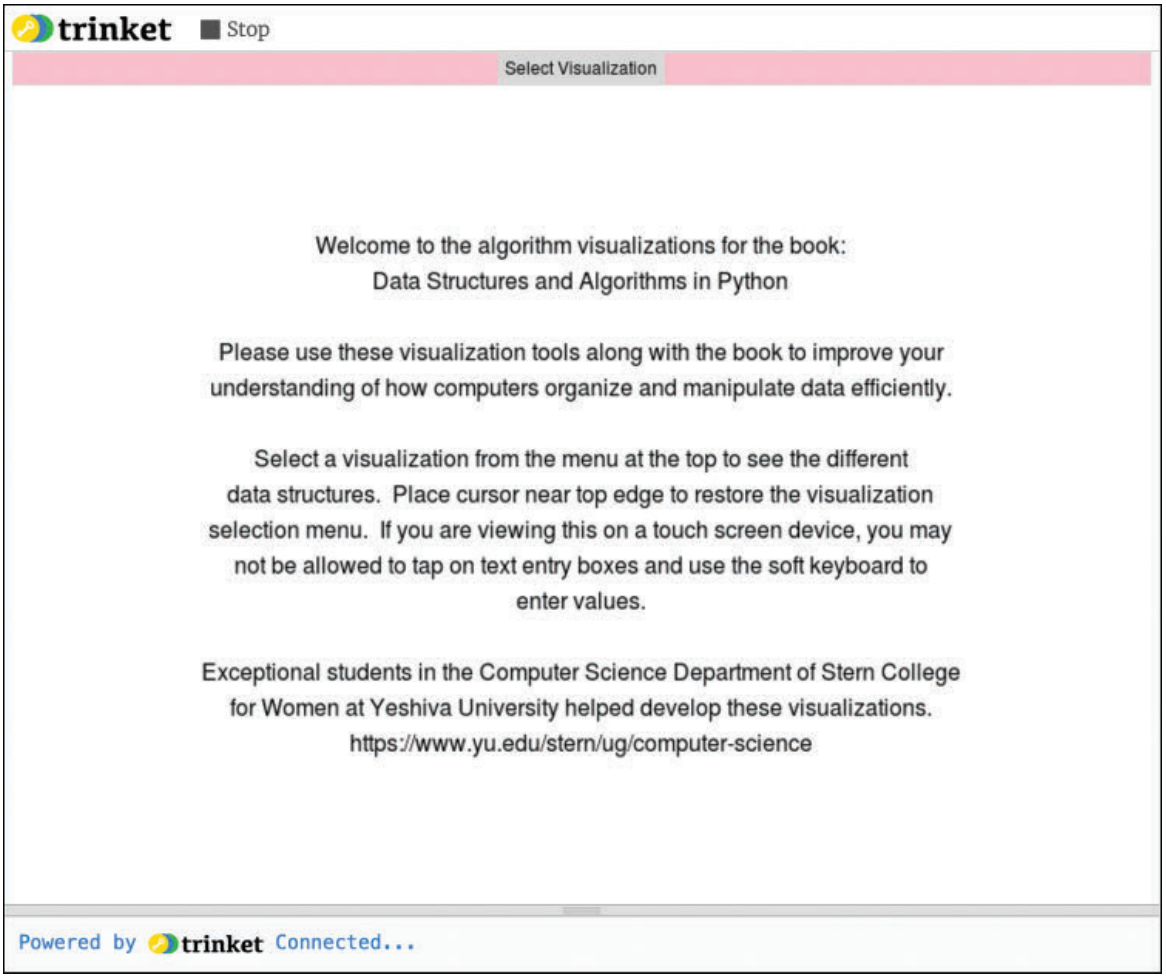

FIGURE A-1 Initial screen when viewing visualizations in a web browser

Select the Select Visualization button at the top to get a menu of all the visualizations that can be shown.

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There are limitations on viewing the visualizations in a web browser. In particular:

- The window cannot be resized although you can increase the size of the upper pane by adjusting the size of the web browser window and dragging the separator bar just above the "Powered by trinket" text.
- The screen is being copied over the Internet, so there might be choppy or slow updates in response to interactions. Try using the visualization tool's lower animation speed or stepping mode when this occurs.
- The viewing works best on computers with a mouse and keyboard. Viewing the trinket.io powered website on a device with only a touchscreen might not allow you to enter text for the arguments to operations. Selecting a text entry area does not put the keyboard focus on that area.

Using the Visualizations

The visualizations are small programs that usually show one data structure at a time. They all have a similar structure with a drawing area at the top and a set of controls on the bottom (see Figure A-2). There are specific instructions for how to use the visualizations in each of the chapters, and some general guidelines common to all of them follow.

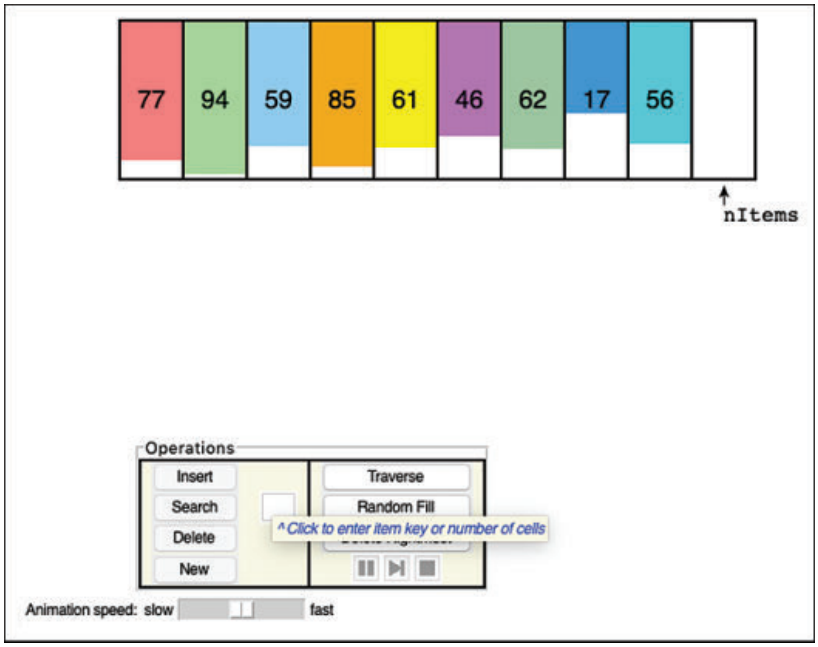

FIGURE A-2 Drawing area at the top and a set of controls on the bottom

The visualizations show the data in a drawing area at the top, such as in Figure A-2. The Operations box in the lower left is where you initiate operations on the data structure. The buttons are grayed out and disabled when they are not applicable. In particular, the operations that take an argument like a key to insert are on the left side of the Operations box. They are grouped with a text entry area that must have a value entered before they become enabled. The hint in blue text explains what can be typed in the text entry area

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for the different operation arguments. Operations that take no arguments, like Traverse and Random Fill, are on the right.

Below the Operations box is a slider. This control can be adjusted to slow down or speed up the animation of the operations. During animations, the three small buttons on the lower right ▶ ▶| ■ are enabled to provide fine control. The leftmost of three buttons is either a triangle to resume playback, or two vertical bars ❚❚ ▶| ■ to pause the animation, depending on the current state. The second button (with a triangle pointing into a vertical bar) steps forward in the animation by one "step." That button is enabled only when code is displayed because the steps are defined by the code. The square button stops the current operation and enables other operations to be run. Stopping an operation before it finishes can leave the data structure in an odd state, such as leaving extra copies of data items or not sorting all the items.

During most operations, a code box appears in the lower right (see Figure A-3). It shows the same program described in the book, highlighting the part of the program being executed. In the figure, the items in an array are being sorted. The various local variables are shown along with the data structure in the drawing at the top. Only the three animation buttons are enabled while the sort operation runs. The highlighting around the single step button indicates that keyboard focus is on that button. Pressing the spacebar when a button has keyboard focus "presses" the button. Operations normally begin running in the "play" mode when a button is pressed. By holding down the Shift key when pressing the button, you can start the operation in stepping mode (if it displays code).

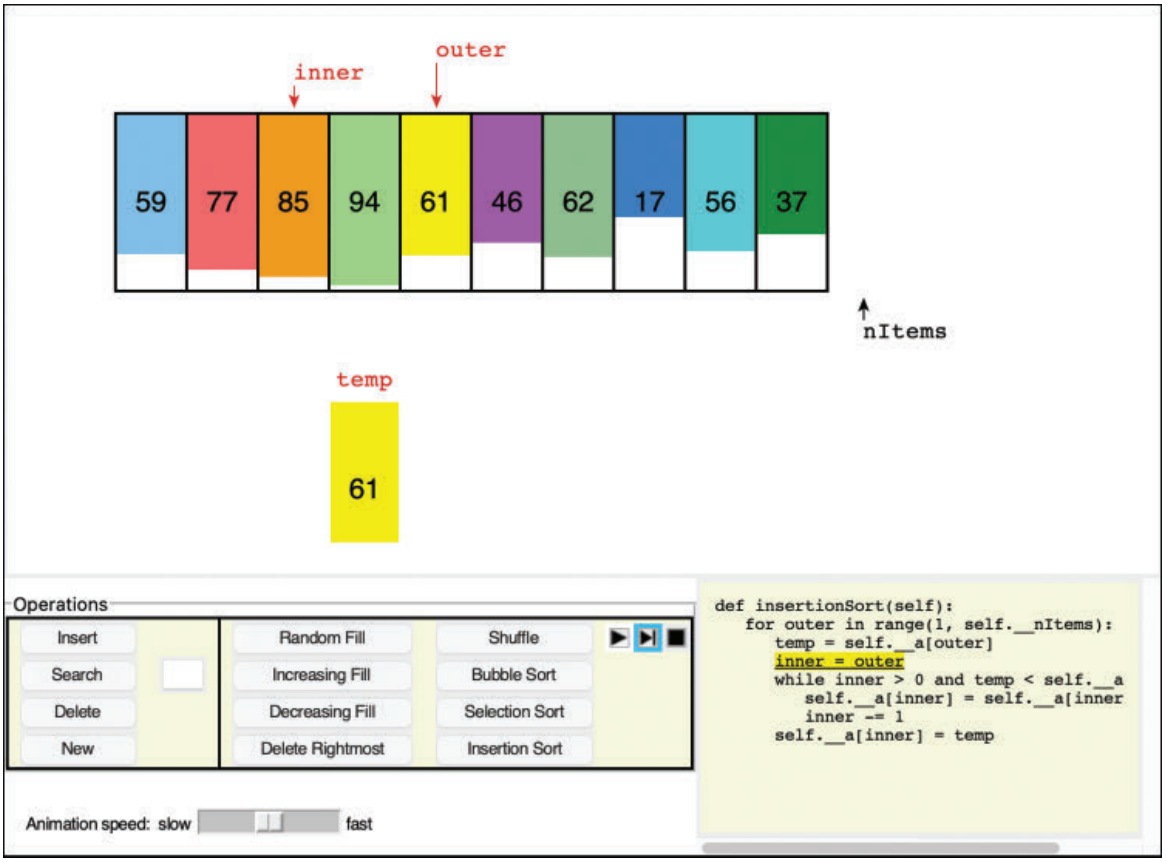

FIGURE A-3 The code box in the lower right

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After each operation finishes, the code and local variables remain on the display until the next operation is started. Status or error messages can appear at the bottom of the window, between the sliding speed control and the code box. The messages usually appear when operations finish. Figure A-4 shows the display after the successful search for key 62 in the array.

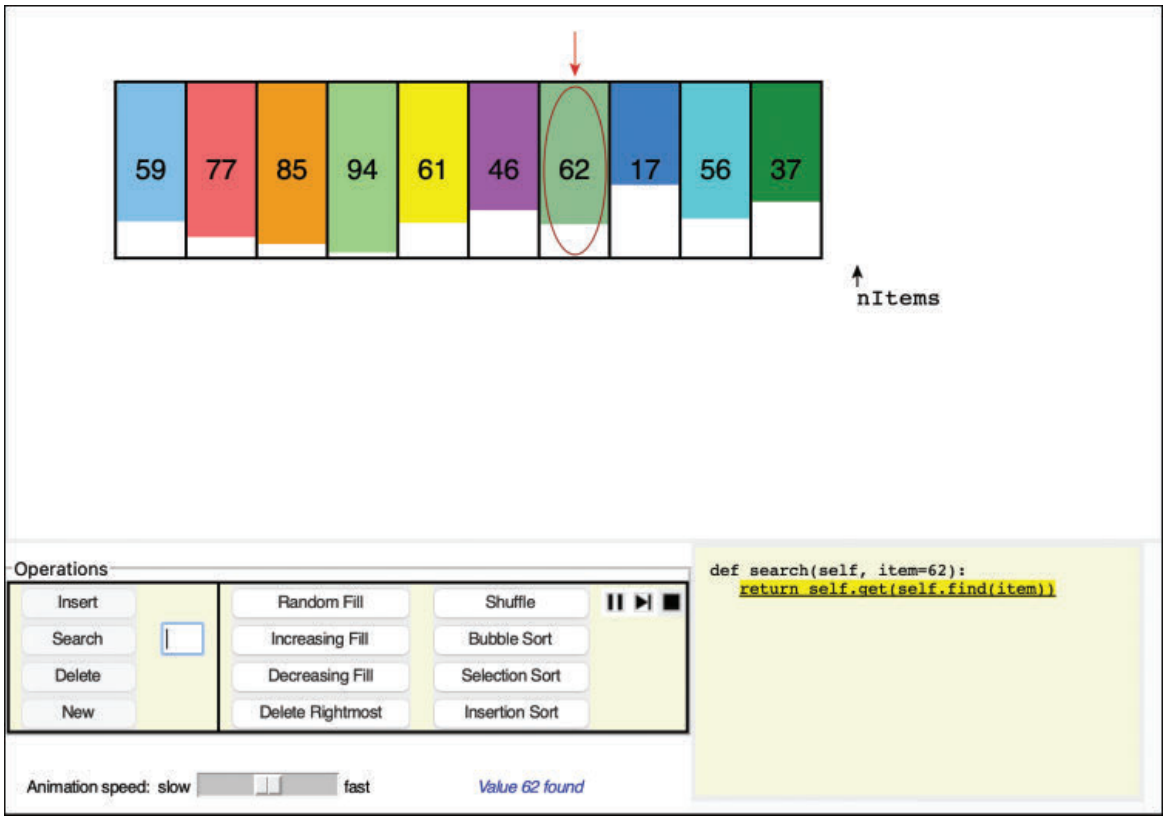

FIGURE A-4 Results of the search for key 62 in the array

In most visualizations, you can click a data item to select it and copy its value into the text entry area. This makes it less likely that a typographical error will prevent searching for or deleting an existing value.

Hint messages about the controls can appear next to the text entry box(es) and operations buttons. Hovering the mouse pointer over one of them displays a short hint about what values can be entered in that box or what the button does. When you first start one of the visualizations, it displays a message telling you all the kinds of things that can be entered in all the text entry box(es), like the message in Figure A-2.

Depending on how you run the visualization, you may be able to resize the window. This capability is sometimes useful when the amount of data grows too large or the lines of code are wide. The code text box does not grow vertically but can be scrolled for longer programs. Some visualizations require more display space and provide scrollbars to move around. At least one visualization also lets users zoom in and out on particular regions.

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